



**Educational Tools
from pre-school to
higher education**

Teaching

Learning

Projects

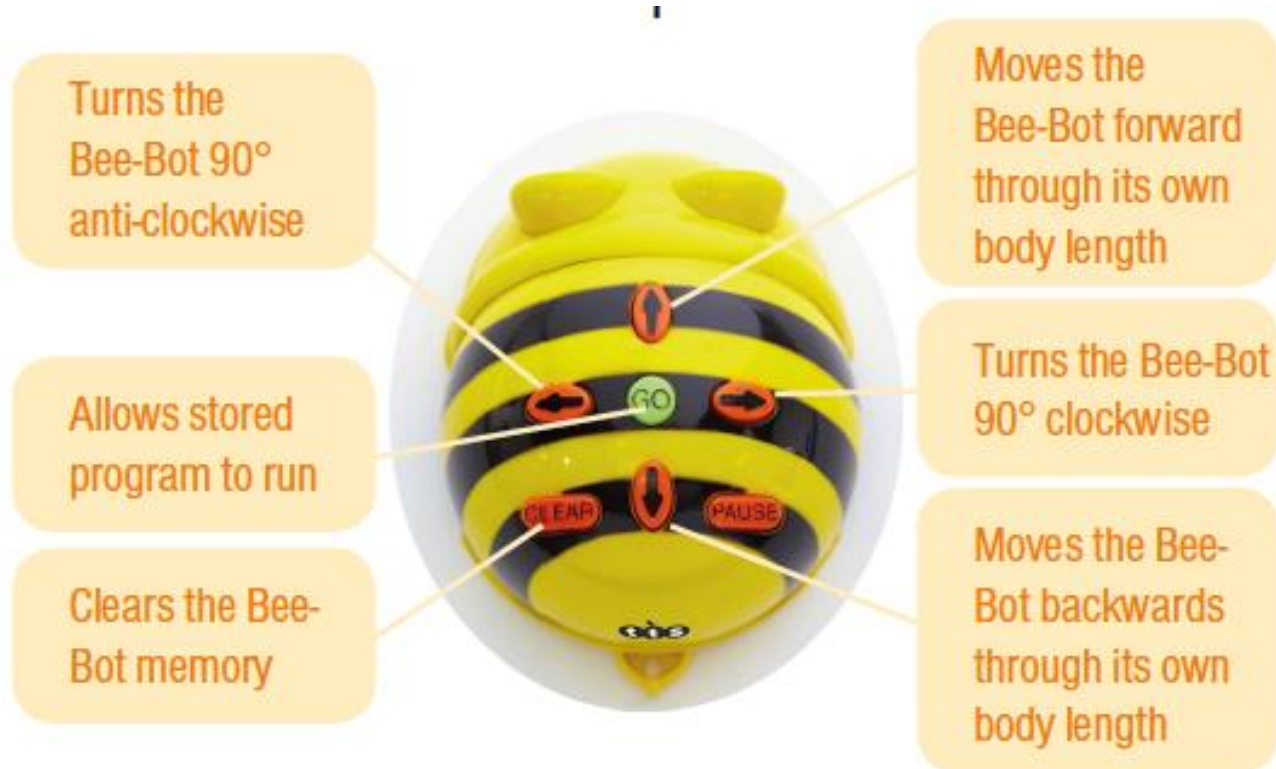
Resources

Anabela Gomes
anabela@isec.pt

CT - Resources for all ages

■ Bee-Bot – 3 Years

■ <https://www.bee-bot.us/>





CT - Resources for all ages

■ Code-a-pillar: 3 Years

- http://www.fisher-price.com/en_US/brands/think-and-learn/index.html





CT - Resources for all ages

5

■ Cubetto (3 years)

■ <https://www.primotoys.com/>





CT - Resources for all ages

■ Kibo (4 years)

- <http://kinderlabrobotics.com/kibo/>





CT - Resources for all ages

7

■ Dash-Dot: 4-12 Years

■ <https://www.makewonder.com/dash>



Wonder



Go



Blockly



Path



Xylo



Swift
Playgrounds™





CT - Resources for all ages

■ Phiro: 4-12 years

- <https://www.kickstarter.com/projects/2074714954/phiro-a-smart-robot-for-kids-learn-to-code-in-5-wa>



PLAY

With endless imagination and creativity, kids play with Phiro and together they go on adventures, create stories, games, explore new worlds, all while having fun.



CODE

As kids play with Phiro, they learn to code and watch Phiro enact their programming. As kids learn to code, they foster computational thinking and problem solving skills, crucial for every field in the 21st century.



INNOVATE

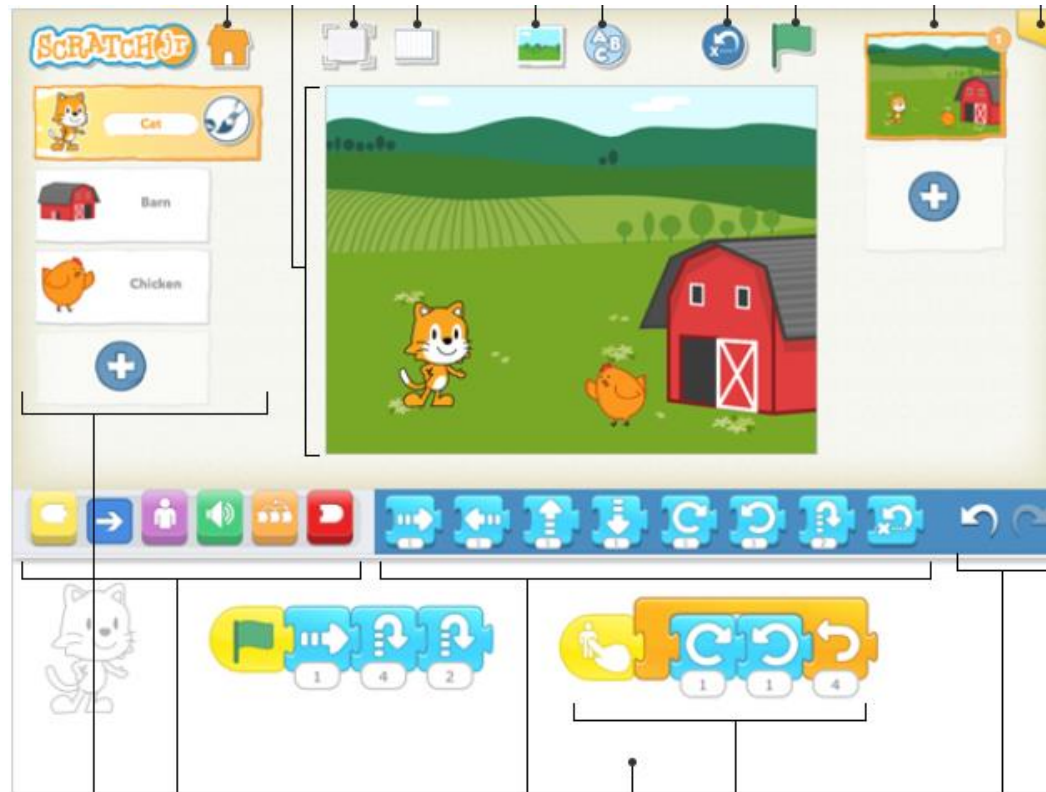
Coding and playing with Phiro empowers and inspires kids to be creators and innovators of the world.



CT - Resources for all ages

■ ScratchJr (5-7 years)

■ <https://www.scratchjr.org/>



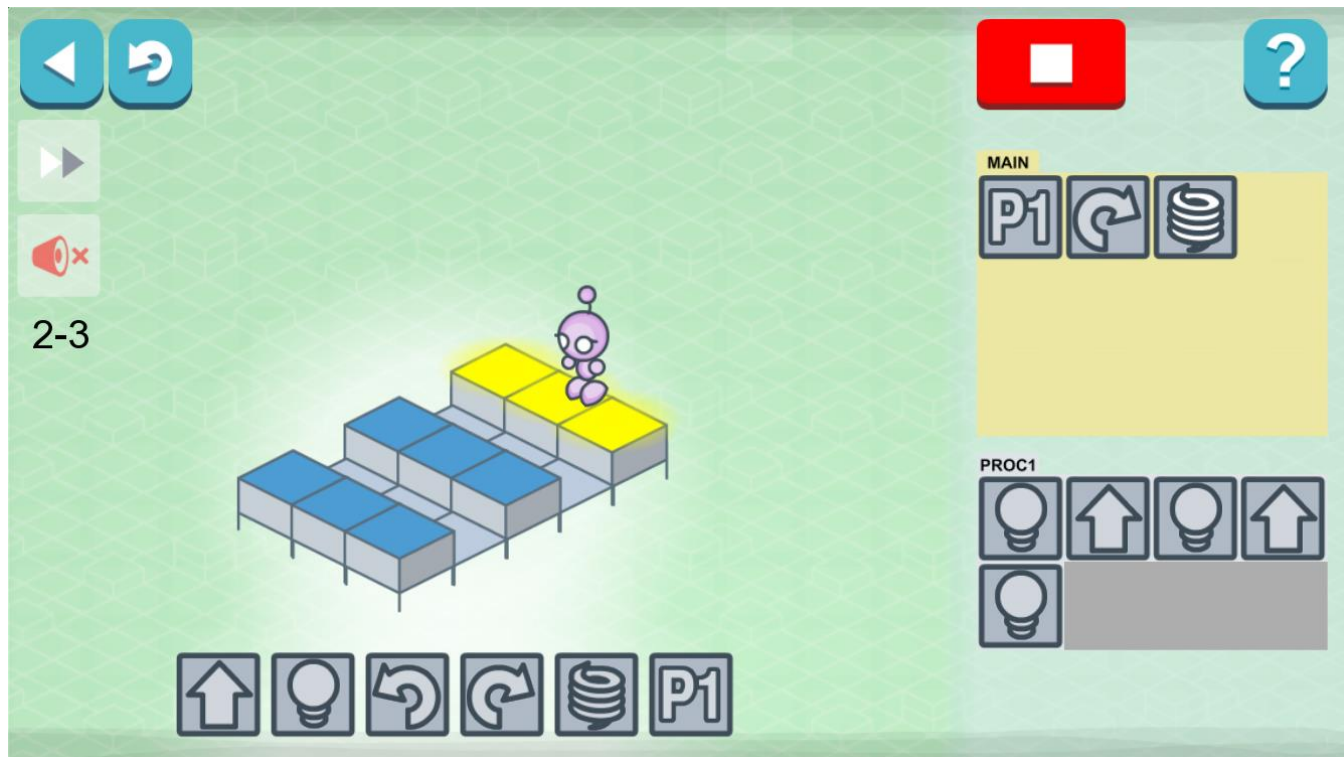


CT - Resources for all ages

10

■ Lightbot

■ <http://lightbot.com/>

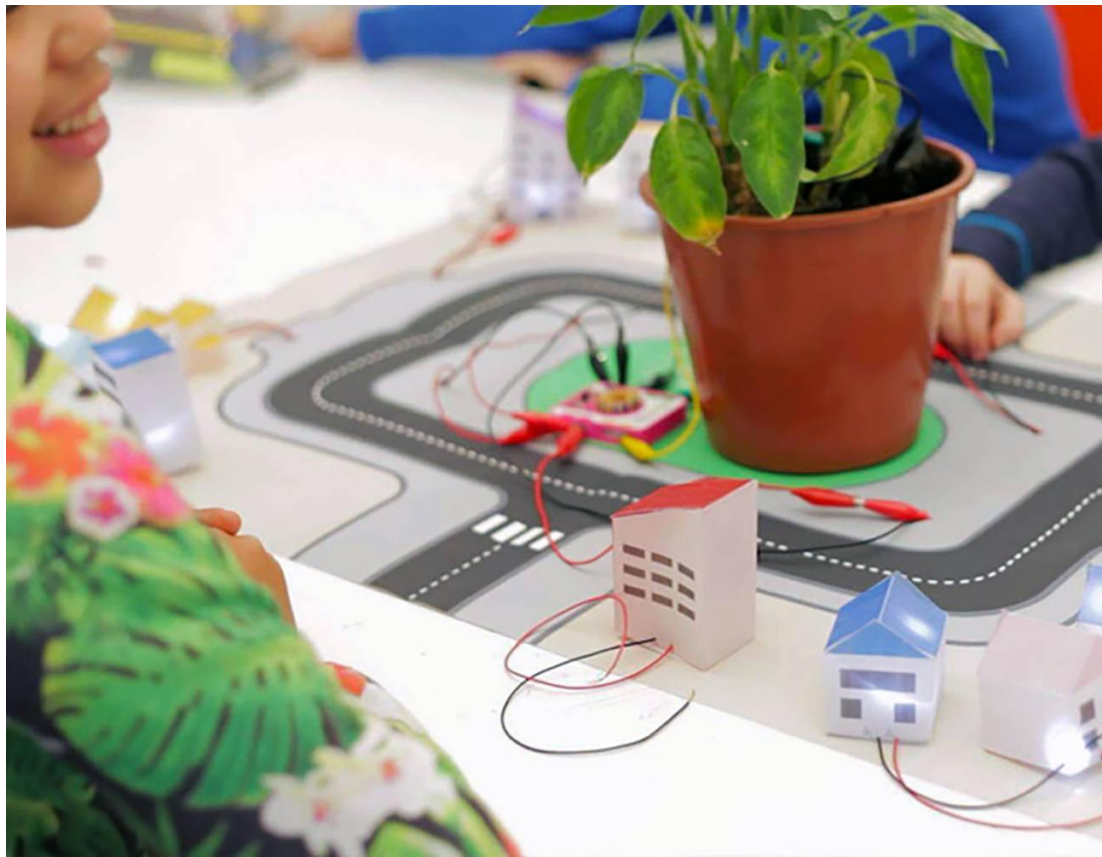




CT - Resources for all ages

11

- QMOD - <https://www.kickstarter.com/projects/qmod/qmod-energy-toys-for-future-innovators?ref=NewsMay0417>



■ Microsoft's Kodu

[illegible]



CT - Resources for all ages

■ MineCraft

- <https://minecraft.net/en-us/>



**FIND LESSON EXAMPLES
ACROSS SUBJECTS AND AGES**

3-5 years

6-7 years

8-10 years

11-13 years

14+ years

Math

Science

Lang Arts

History

Visual Art

CT - Resources for all ages

■ Scratch

■ <https://scratch.mit.edu/>

[Create](#)[Explore](#)[Tips](#)[About](#)[Join Scratch](#)[Sign in](#)

Create stories, games, and animations
Share with others around the world



A creative learning community with **25,064,069** projects shared



CT - Resources for all ages

■ MicroWorlds EX (LOGO)

- <http://www.microworlds.com/>





CT - Resources for all ages

16

■ TurtleAcademy (LOGO)

■ <https://turtleacademy.com/>



5. Cool labels6. Loops7. Polygons8. The pen width

lesson8: The pen width

1. The Width of the Pen

Until now you have been using a pen that draws a black line the width of 1 point. The width of the line means how thick the line is. If we want to draw more beautiful things, sometimes we'll want to use a wider or narrower line, or choose a different color. The command to change the pens width is `setwidth` followed by a number. The number will represent the new width of the line, counting it in points. Set the pen width to 5.

Solution

2. A Wider Line

3. Even Wider

4. Taking a Diagonal Turn

5. A Line with Changing Width

```
> seth 0
> penup
> fd 10
> fd 10
> seth 90
> pendown
> fd 20
> fd 10
> fd 5
> penup
> hideturtle
>
```

←Prev→Next

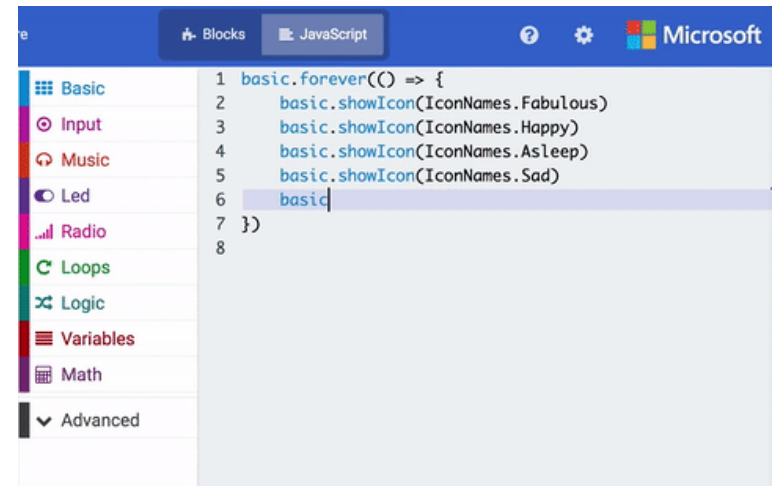
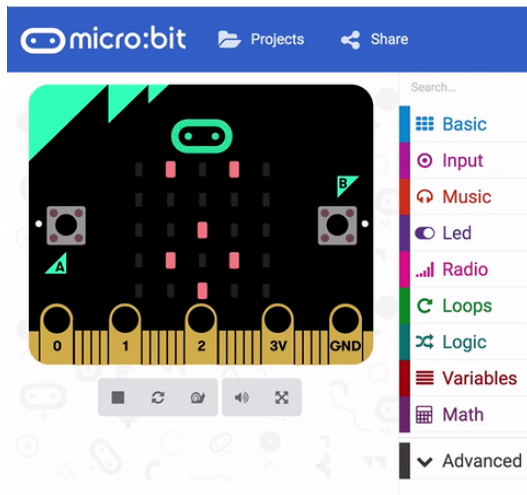


CT - Resources for all ages

17

■ TouchDevelop -> MakeCode

■ <https://www.touchdevelop.com/>

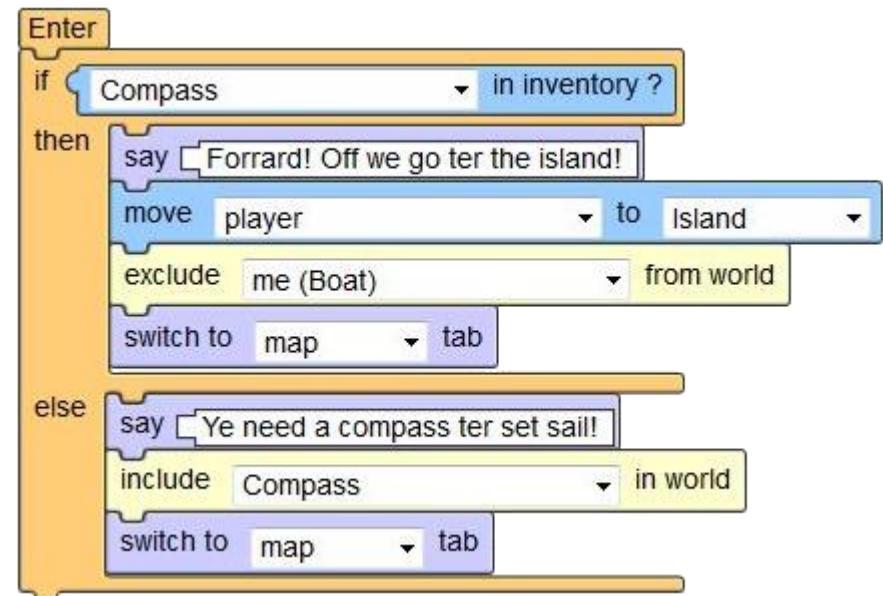
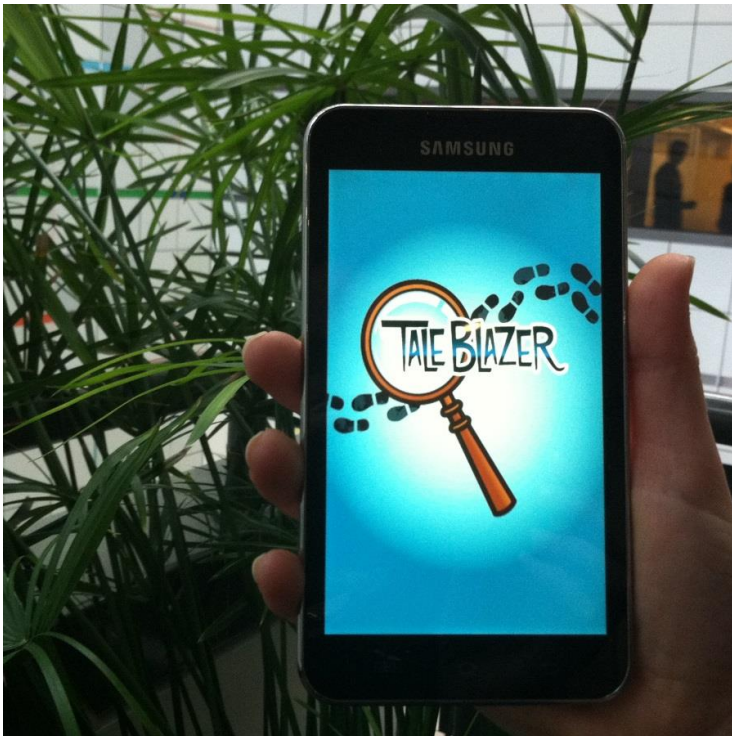




CT - Resources for all ages

■ TaleBlaze

■ <http://taleblazer.org/>



- App Inventor
 - <http://appinventor.mit.edu>

